

Syllabus for Geotechnical Engineering

Welcome to Geotechnical Engineering! In this course, we will explore what lies beneath the Earth's surface. We will study the properties of soils found around the world, how those properties influence engineering behavior under different conditions, and how they are measured and classified. Through lectures, we will examine soil mechanics concepts and learn how to calculate, analyze, and interpret geotechnical data. In the laboratory, we will gain hands-on experience investigating soil samples, determining soil classifications, and evaluating key properties introduced in lecture.

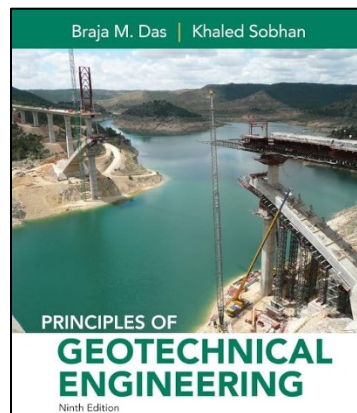
INSTRUCTOR

Yingxiao Liu, Ph.D (liuyin@rowan.edu)

Engineering Hall, Room 227

TEXTBOOK

Principles of Geotechnical Engineering, Braja M. Das, 9th edition



Older editions are also acceptable. Additionally, supplemental sources such as journal articles, news articles, and sections of various literature will be distributed to you.

PREREQUISITES

A grade of **C- or higher** in **Civil Engineering Materials**, **Fluid Mechanics**, and **Solid Mechanics** is required to enroll in this course. Students are expected to have retained and be comfortable applying the fundamental concepts from these courses to succeed in Geotechnical Engineering. In particular, this course builds upon prior knowledge of particle size analysis and material

classification, fluid flow and Bernoulli's equation, stress-strain behavior of materials, and the packing and interaction of particulate media.

This course is a prerequisite for Foundation Engineering and Earth Retaining Systems. A grade of C- or higher in Geotechnical Engineering is required to enroll in these subsequent courses.

FINAL GRADING BREAKDOWN

- Homework and Classwork 15%
- Quizzes 20%
- Lab participation and reports 20%
- Midterm 20%
- Final Exam (cumulative) 25%

Extra credit opportunities will be offered during the semester to enhance student learning. These opportunities will be open to all students and will account for no more than 15% of the total course grade.

ATTENDANCE

In accordance with university policy (see p. 28 of the 1995–96 Rowan Undergraduate Catalog), attendance is required because substantial course content is delivered during class meetings. Students are expected to attend **all lectures and laboratory sessions** throughout the semester. Each student is permitted **two (2) unexcused absences** without penalty. A third unexcused absence will result in the loss of a **sign grade** (e.g., B+ to B), a fourth will result in the loss of a **full letter grade** (e.g., B+ to C+), and **five (5) unexcused absences will result in an automatic failure of the course.**

- Absences due to **religious observances, official university activities, documented illness, family emergency (including death or serious injury of a family member or loved one), or inclement weather** will be considered excused. Students must notify the instructor and provide **appropriate written documentation** prior to the absence or as soon as reasonably possible.
- A **5% bonus** will be applied to the final course grade of students who maintain **full attendance** for the entire semester.
- If a student is unable to attend a scheduled laboratory session, they must arrange a **make-up lab** with the instructor in advance and complete the missed experiment in another lab section, subject to availability.

QUIZZES

Several quizzes will be given throughout the semester, typically **one per chapter covered**. Additional quizzes may be given as needed, depending on course content and pacing. All quizzes will be completed **individually**. The **lowest quiz score will be dropped** when calculating the final quiz average.

For each quiz, students may bring one letter-sized double-sided cheat sheet, containing equations or notes. **No electronic devices are permitted during quizzes except for calculators.**

EXAMS

There will be **two exams** in this course: a **midterm exam** and a **final exam**. The midterm exam will be administered after the completion of the **Compaction of Soils** topic, and the final exam will be held during **finals week**, as scheduled by the university.

For each exam, students may bring one letter-sized double-sided cheat sheet, containing equations or notes. **No electronic devices are permitted during quizzes except for calculators.**

LAB

Several laboratory sessions will be conducted throughout the semester. The specific weeks in which laboratories are held will be announced by the instructor. Students are expected to **read and understand the laboratory procedures before** each experiment. If the instructor determines that students are not adequately prepared, **weekly laboratory quizzes** may be administered and will count toward **individual grades**.

There will be **four (4) laboratory reports** during the semester, as outlined in the course schedule below. Each report is due **one week after completion of the final experiment** in the corresponding laboratory set. Detailed **guidelines for laboratory report submissions** will be provided prior to the first submission.

Laboratory groups will be **assigned by the instructor**.

As part of the laboratory grade, a **group project** may be assigned toward the end of the semester. The project will consist of a **written paper and an oral presentation**, potentially focused on a geotechnical engineering issue or failure case from the past ten years. Additional details, including scope, expectations, and grading criteria, will be provided later in the semester.

INCLUSIVE LEARNING ENVIRONMENT

I consider this classroom to be a space in which all students are treated with respect, dignity, and fairness. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities and expressions, national origins, religious affiliations, sexual orientations, socioeconomic backgrounds, family educational histories, abilities, and other visible and non-visible differences. The diversity of perspectives and experiences that students bring to this course is valued and viewed as an asset to our shared learning.

All members of this class are expected to contribute to a respectful, welcoming, and inclusive learning environment. If at any time you feel that your contributions are not being valued or that you are not being treated respectfully, I encourage you to speak with me privately. Anonymous communication is also welcome. In addition, students may contact the Office of Social Justice, Inclusion, and Conflict Resolution (socialjustice@rowan.edu, 856-256-5496, Room 118, Robinson Hall) for support.

My intention is that students from all backgrounds and perspectives are well served by this course, and I welcome your feedback and suggestions as we work together to maintain an inclusive and supportive learning environment.

SPECIAL ACCOMMODATION

Your academic success is important. Students with a documented disability that may affect their participation or performance in this course are encouraged to contact the instructor as early as possible. To receive official university accommodations, students must provide appropriate documentation to the Academic Success Center.

The Academic Success Center is available to answer questions and assist students in obtaining accommodations. The Center can be reached at 856-256-4234 and is located on the third floor of Savitz Hall. We look forward to working with you to support your learning goals.

ACADEMIC INTEGRITY POLICY

Each student should become familiar with Rowan University's Academic Integrity Policy, including the types of violations and subsequent penalties. Violations of the academic integrity policy include any form of plagiarism, which occurs when copying information or ideas from published and unpublished sources without correctly citing the source. Additionally, using generative AI (such as Chat GPT, Claude CoPilot, Gemini, & Grammarly) without instructor permission, accessing prohibited resources during assignments, receiving unauthorized assistance, and posting course materials online are all considered violations. Students committing any act of academic dishonesty may fail the assignment, fail the course, or face other sanctions. Please work with your instructor on any questions regarding how academic integrity applies in a given course.

or assignment, and please review the resources available at [go.rowan.edu/academic integrity](https://go.rowan.edu/academic-integrity). Ensuring that your learning is genuine is the key to your future success and your integrity, and is a core value of the Rowan community.

TIPS FOR SUCCESS

- **Check your email often for updates and announcements.**
 - Get to know your classmates and form study groups.
 - Come to class regularly and be prepared.
 - Take homework seriously.
 - Ask for help early if something doesn't make sense.
 - Meet with your advisor at least once each semester.
 - Manage your time well and stay organized.
 - Get involved in extracurricular or service activities when you can.
 - Stay curious, keep up with the world, and enjoy learning.
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TENTATIVE COURSE SCHEDULE

Topic	Objective	Lab
Soil Basics	Learn the soil origins, mineralogy, specific gravity, and grain sizes.	Lab 0-Safety Training (week 1) Lab 1- Specific Gravity and Particle Size Analysis (week 4)
Weight-Volume Relationships	Acquire a solid knowledge of how certain soil properties, such as void ratio and moisture content, affect a soil's behavior.	
Plasticity and Structure	Learn the basics of the Liquidity Index. Characterize and classify soils. Understand the relationship between soil packing and fabric.	Lab 2- Atterberg Limits (week 5)
Soil Classification	Understand the methods and the process of the importance of soil classifications through USCS, AASHTO, and USGS Textural guidelines.	Report 1- Classification of Soil (week 6)
Compaction of Soils	Understand the principles and methods of field compaction and how the field results are dictated and analyzed through the analysis of laboratory work.	Lab 3- Compaction Test Sand Cone Test (week 8) Report 2- Soil Compaction (week 10)
Soil Permeability	Develop an understanding of how fluid mechanics affects soils and vice versa. Investigate which properties influence more or less permeability, and how to analyze and calculate numerical relationships	Lab 4- Constant Head Permeability Test (week 11)
Seepage	Discover the importance of Flow Nets, how they are quantified and qualified, and learn how to compute important analysis which pertains to seepage through, around, and under structures.	Lab 5- Flow Net Computation (week 12) Report 3- Permeability and Seepage (week 13)
Soil Stresses	Learn how to calculate In Situ stresses and stresses from the application of loads to the surface. Master the Mohr's Circle.	
Soil Compression	Learn different types of settlement and consolidation, as well as discuss the reasons behind the phenomena and the properties and methods of quantifying them.	
Shear Strength of Soils	Investigate and discuss the methods of quantifying shear strength, the differences in methods, and the computational tools needed to determine the strength of a soil	Final presentation (week 14)

Note: The schedule of topics and assignments is subject to change. The **laboratory sequence and scheduling** will be determined by the instructor.